Complex Numbers: De Moivre's Theorem - Multiple Angles

Questions 1–23

Question 1. By first expanding the brackets using a binomial expansion, find each of the following in its simplest form.

a) $(1+\mathrm{i})^3$
b) $(1+\mathrm{i})^4$
c) $(1+\mathrm{i})^5$
d) $(1+\mathrm{i})^6$

Question 2. By first expanding the brackets using a binomial expansion, find each of the following in its simplest form in terms of a and b where a and b are positive real constants.

a) $(a+b\mathrm{i})^3$
b) $(a+b\mathrm{i})^4$
c) $(a+b\mathrm{i})^5$

Question 3.

a) Find the expansion of $(\cos\theta + \mathrm{i}\sin\theta)^2$

b) Hence show that $\cos 2\theta = \cos^2\theta - \sin^2\theta$ and $\sin 2\theta = 2\sin\theta\cos\theta$

Question 4.

a) Find the expansion of $(\cos\theta + \mathrm{i}\sin\theta)^3$

b) Hence show that

$$\cos 3\theta = 4\cos^3\theta - 3\cos\theta \text{ and } \sin 3\theta = 3\sin\theta - 4\sin^3\theta$$

c) Hence find $\tan 3\theta$ in terms of powers of $\tan\theta$ only.

Question 5. Given $\cos 3\theta = 4\cos^3\theta - 3\cos\theta$

a) Solve $4x^3 - 3x = 1$

b) Find the solutions to $\cos 3\theta = 1$ for $0 \leq \theta \leq \pi$

c) Hence, by considering $x = \cos\theta$, verify all the solutions to $4x^3 - 3x = 1$ in the interval $-1 \leq x \leq 1$

Question 6.

a) Use De Moivre's theorem to show that

$$\cos 4\theta = 8\cos^4\theta - 8\cos^2\theta + 1$$

b) By using $\cos 4\theta = -1$, find the exact values to the equation $x^4 - x^2 = -\frac{1}{4}$

Prerequisite knowledge for Q7: Reversing the Chain Rule

Question 7.

a) Show that

$$\sin 4\theta = 4\cos^3\theta\sin\theta - 4\cos\theta\sin^3\theta$$

b) Hence show that

$$\int_0^{\frac{3\pi}{4}} (\cos^3\theta\sin\theta - \cos\theta\sin^3\theta)\,\mathrm{d}\theta = \frac{1}{8}$$

Question 8. Use De Moivre's theorem to show that

$$\cot 3\theta = \frac{\cot^3\theta - 3\cot\theta}{3\cot^2\theta - 1}$$

Question 9.

a) Show that $\cos 5\theta = 16\cos^5\theta - 20\cos^3\theta + 5\cos\theta$

b) Show that $\sin 5\theta = 16\sin^5\theta - 20\sin^3\theta + 5\sin\theta$

c) Find $(1 + \tan^2\theta)^2$

d) Hence, show that

$$\tan 5\theta = \frac{\tan^5\theta - 10\tan^3\theta + 5\tan\theta}{1 - 10\tan^2\theta + 5\tan^4\theta}$$

e) Find the five distinct real solutions to

$$1 - 10x^2 + 5x^4 = x^5 - 10x^3 + 5x$$

giving your answers to three decimal places.

Question 10.

a) Find $(1 - \cos^2\theta)^3$

b) Show that

$$\cos 6\theta = 32\cos^6\theta - 48\cos^4\theta + 18\cos^2\theta - 1$$

c) Hence, find the six solutions to

$$64x^6 - 96x^4 + 36x^2 - 3 = 0$$

giving your answers to four decimal places.

Question 11. Show that

$$\sin 6\theta = a\cos^5\theta\sin\theta - b\cos^3\theta\sin^3\theta + c\cos\theta\sin^5\theta$$

where a, b and c are integers to be found.

Question 12.

a) Find $\frac{\sin 4\theta}{\sin\theta}$ in terms of $\cos\theta$

b) Hence solve the equation

$$\sin 4\theta = \sin\theta$$

for $0 < \theta < \pi$ giving your answers to three significant figures.

Question 13.

a) Find $\frac{\cos 5\theta}{\cos \theta}$ in terms of $\cos \theta$

b) Hence solve the equation

$$\cos 5\theta = -\cos \theta$$

for $0 < \theta \leq \pi$ giving your answers in exact form.

Question 14. It is given that $z = 1 + 3\mathrm{i}$

a) Find z^3 in the form $a + b\mathrm{i}$ where a and b are real numbers.

b) Let $\theta = \arg z$. Find $\arg z$ giving your answer in exact form.

c) Use De Moivre's theorem to show that

$$\cos 3\theta = 4\cos^3 \theta - 3\cos \theta \text{ and } \sin 3\theta = 3\sin \theta - 4\sin^3 \theta$$

d) Show that

$$\tan 3\theta = \frac{3\tan \theta - \tan^3 \theta}{1 - 3\tan^2 \theta}$$

e) Use your results to verify that the value of $\tan \theta$ obtained from z is consistent with the identity for $\tan 3\theta$

Prerequisite Knowledge for Q15-Q18: Hyperbolic Functions and Euler's Formula

Question 15.

a) Expand $(\cosh x + \sinh x)^3$

b) Show that

$$(\cosh x + \sinh x)^3 = \cosh 3x + \sinh 3x$$

and $(\cosh x - \sinh x)^3 = \cosh 3x - \sinh 3x$

c) Hence, find $\cosh 3x$ and $\sinh 3x$ in terms of $\cosh x$ and $\sinh x$

d) Given that $\cosh x = 2$ and $\sinh x = \sqrt{3}$, determine the exact value of $\cosh 3x$ and $\sinh 3x$

Question 16.

a) By using Euler's formula show that $\cosh(\mathrm{i}z) = \cos z$

b) Hence, find an exact value for $\cosh\left(\frac{3\pi}{4}\mathrm{i}\right)$

Question 17.

a) By using Euler's formula show that $\sinh(\mathrm{i}z) = \mathrm{i}\sin z$

b) Hence, find an exact value for $\sinh\left(\frac{5\pi}{6}\mathrm{i}\right)$

Question 18. Given $\sinh(\mathrm{i}z) = \mathrm{i}\sin(z)$, show that

$$\sinh\left(\frac{3\pi}{7}\mathrm{i}\right) = 3\mathrm{i}\cos^2\left(\frac{\pi}{7}\right)\sin\frac{\pi}{7} - \mathrm{i}\sin^3\left(\frac{\pi}{7}\right)$$

Question 19. Use De Moivre's theorem to find $\cos 3\theta$ and hence show that

$$x = \cos\left(\frac{4\pi}{9}\right)$$

satisfies $8x^3 - 6x + 1 = 0$

Question 20.

a) Show that $\cos 5\theta = 16\cos^5\theta - 20\cos^3\theta + 5\cos\theta$

b) By considering the solutions of $\cos(5\theta) = 0$ for $0 \leq \theta \leq \pi$, show that

$$\cos^2\left(\frac{\pi}{10}\right) = \frac{5+\sqrt{5}}{8}$$

and

$$\cos^2\left(\frac{3\pi}{10}\right) = \frac{5-\sqrt{5}}{8}$$

Question 21.

a) Show that

$$\cos 6\theta = 32\cos^6\theta - 48\cos^4\theta + 18\cos^2\theta - 1$$

b) By considering the solutions of $\cos 6\theta = 0$ for $0 \leq \theta \leq \pi$, show that

$$\cos^2\left(\frac{\pi}{12}\right) = \frac{2+\sqrt{3}}{4}$$

and

$$\cos^2\left(\frac{5\pi}{12}\right) = \frac{2-\sqrt{3}}{4}$$

Question 22.

a) Use De Moivre's theorem to find $\cos 5\theta$

b) By considering $\cos 5\theta = 1$, show that x satisfies the equation

$$(x-1)(4x^2+2x-1)^2 = 0$$

c) Hence, show that

$$\cos 72^\circ = \frac{-1+\sqrt{5}}{4}$$

and

$$\cos 144^\circ = \frac{-1-\sqrt{5}}{4}$$

Prerequisite Knowledge for Q23: Geometric Series

Question 23.

a) Find $\cos n\theta$ in terms of $\sin \frac{n\theta}{2}$

b) Find $\sin n\theta$ in terms of $\sin \frac{n\theta}{2}$ and $\cos \frac{n\theta}{2}$

c) Show that

$$\sum_{r=1}^{n} \cos(r\theta) = \frac{\cos\left(\frac{(n+1)\theta}{2}\right)\sin\left(\frac{n\theta}{2}\right)}{\sin\left(\frac{\theta}{2}\right)}$$

d) Show that

$$\sum_{r=1}^{n} \sin(r\theta) = \frac{\sin\left(\frac{(n+1)\theta}{2}\right)\sin\left(\frac{n\theta}{2}\right)}{\sin\left(\frac{\theta}{2}\right)}$$